

Section 2

Mechanism Design Basics
(the science of rule-making)

The design of single-item auctions

Course Goal 1: Understand how to design systems with strategic participants that have good performance guarantees.

Example: Consider a seller with a single item, such as a slightly antiquated smartphone. This is the setup in a typical eBay auction, for example. There is some number n of (strategic!) bidders who are potentially interested in buying the item.

- Bidder behavior in various auction formats
- model of what a bidder wants

The design of single-item auctions

- Each bidder i
 - a nonnegative valuation v_i
(her maximum willingness-to-pay for the item being sold)
 - Thus bidder i wants to acquire the item as cheaply as possible, provided the selling price is at most v_i .
 - Another important assumption is that this valuation is private, meaning it is unknown to the seller and to the other bidders.
- quasilinear utility model:
 - If a bidder i loses an auction, her utility is 0.
 - If the bidder wins at a price p , her utility is $v_i - p$.

Sealed-Bid Auctions

- 1) Each bidder i privately communicates a bid b_i to the seller, in a sealed envelope, if you like.
- 2) The seller decides who gets the item (if anyone).
- 3) The seller decides on a selling price.
 - There is a way to implement the 2nd step, **give the item to the highest bidder**.
 - There are multiple reasonable ways to implement the 3rd step, and the choice of implementation significantly affects bidder behavior. For example, suppose we try to be altruistic and charge the winning bidder nothing. This idea backfires badly, with the auction devolving into a game of “who can name the highest number?”

First-Price Auctions

In a first-price auction, the winning bidder **pays her bid**.

First-price auctions are **hard** to reason about.

- First, as a participant, it's hard to figure out **how to bid**.
- Second, as a seller or auction designer, it's hard to predict **what will happen**.

Second-Price Auctions and Dominant Strategies (1)

What happens when you win an eBay auction?

If you bid \$100 and win, do you pay \$100?

Not necessarily: eBay uses a “proxy bidder” that increases your bid on your behalf until your maximum bid is reached, or until you are the highest bidder, whichever comes first.

For example, if the highest other bid is only \$90, then you only pay \$90 (plus a small increment) rather than your maximum bid of \$100. If you win an eBay auction, the sale price is essentially the highest other bid—the **second highest overall**.

Second-Price Auctions and Dominant Strategies (2)

A **second-price** or **Vickrey** auction is a sealed-bid auction in which the highest bidder wins and pays a price equal to the second-highest bid. To state the most important property of second-price auctions, we define a **dominant strategy** as a strategy (i.e., a bid) that is guaranteed to maximize a bidder's utility, no matter what the other bidders do.

Second-Price Auctions and Dominant Strategies (3)

Proposition 2.1 (Incentives in Second-Price Auctions) In a second-price auction, every bidder i has a dominant strategy: set the bid b_i equal to her private valuation v_i .

Proposition 2.1 **implies** that second-price auctions are particularly easy to participate in. When selecting a bid, a bidder doesn't need to reason about the other bidders in any way—how many there are, what their valuations are, whether or not they bid truthfully, etc.

This is completely different from a first-price auction, where it never makes sense to bid one's valuation—this guarantees zero utility—and the optimal amount to underbid depends on the bids of the other bidders.

Second-Price Auctions and Dominant Strategies (4)

Another important property is that a truthful bidder—meaning one that bids her true valuation—never **regrets** participating in a second-price auction.

Proposition 2.2 (Nonnegative Utility) In a second-price auction, every truthful bidder is guaranteed nonnegative utility.

Ideal Auctions (1)

Second-price single-item auctions are “ideal” in that they enjoy **three** quite different and **desirable** properties.

Definition 2.3 (Dominant-Strategy Incentive Compatible) An auction is dominant-strategy incentive compatible (DSIC) if truthful bidding is always a dominant strategy for every bidder and if truthful bidders always obtain nonnegative utility.

(*a.k.a.* individual rationality or voluntary participation.)

Ideal Auctions (2)

Define the social welfare of an outcome of a single-item auction by

$$\sum_{i=1}^n v_i x_i,$$

sale price does not appear in the definition of the social welfare?

where x_i is 1 if i wins and 0 if i loses. Because there is only one item, we have the feasibility constraint that $\sum_{i=1}^n x_i \leq 1$. Thus, the social welfare is just the valuation of the winner, or 0 if there is no winner. An auction is welfare maximizing if, when bids are truthful, the auction outcome has the maximum possible social welfare.

Ideal Auctions (3)

Theorem 2.4 (Second-Price Auctions Are Ideal) A second price single-item auction satisfies the following:

- (1) [*strong incentive guarantees*] It is a DSIC auction.
- (2) [*strong performance guarantees*] It is welfare maximizing.
- (3) [*computational efficiency*] It can be implemented in time polynomial (indeed, linear) in the size of the input, meaning the number of bits necessary to represent the numbers $v_1, \dots, v_n$.

Case Study: Sponsored Search Auctions

A Web search results page comprises a list of organic search results—deemed relevant to your query by an algorithm like *PageRank*—and a list of sponsored links, which have been paid for by advertisers.

Every time you type a search query into a search engine, an auction is run in real time to decide which advertisers' links are shown, how these links are arranged visually, and what the advertisers are charged.

Around 2006, sponsored search auctions generated roughly 98% of Google's revenue.

The Basic Model of Sponsored Search Auctions (1)

The items for sale are k “slots” for sponsored links on a search results page. The bidders are the advertisers who have a standing bid on the keyword that was searched on. For example, **Volvo and Subaru** might be bidders on the keyword “station wagon,” while **Nikon and Canon** might be bidders on the keyword “camera.”

The Basic Model of Sponsored Search Auctions (2)

We quantify the difference between different slots using *click through rates* (CTR_s). The CTR α_j of a slot j represents the probability that the end user clicks on this slot. Ordering the slots from top to bottom, we make the reasonable assumption that $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_k$. For simplicity, we also make the unreasonable assumption that the CTR of a slot is independent of its occupant. Everything we'll say about sponsored search auctions extends to the more general and realistic model in which each advertiser i has a “quality score” β_i (the higher the better) and the CTR of advertiser i in slot j is the product $\beta_i \alpha_j$ (e.g., Exercise 3.4).

The Basic Model of Sponsored Search Auctions (3)

We assume that an advertiser is not interested in an impression (i.e., being displayed on a page) per se, but rather has a private valuation v_i for each click on her link. Hence, the expected value derived by advertiser i from slot j is $v_i \alpha_j$.

The Basic Model of Sponsored Search Auctions (4)

What We Want (desiderata):

- (1) **DSIC**. That is, truthful bidding should be a dominant strategy, and never leads to negative utility.
- (2) **Social welfare maximization**. That is, the assignment of bidders to slots should maximize $\sum_{i=1}^n x_i v_i \leq 1$, where x_i now denotes the CTR of the slot to which i is assigned (or 0 if i is not assigned to a slot). Each slot can only be assigned to one bidder, and each bidder gets only one slot.
- (3) **Computational efficiency**. The running time should be polynomial (or even near-linear) in the size of the input $v_1, \dots, v_n$. Remember that zillions of these auctions need to be run every day!

The Basic Model of Sponsored Search Auctions (5)

Our Design Approach

(1) the choice of who wins what, and (2) the choice of who pays what.

Step 1: Assume, without justification, that bidders bid truthfully. Then, how should we assign bidders to slots so that the above properties (2) and (3) hold?

Step 2: Given our answer to Step 1, how should we set selling prices so that the above property (1) holds?

The Basic Model of Sponsored Search Auctions (6)

Our Design Approach

(1) the choice of who wins what, and (2) the choice of who pays what.

Step 1: the natural greedy algorithm is optimal (and computationally efficient): for $i = 1, 2, \dots, k$, assign the i -th highest bidder to the i -th best slot.

Step 2: there is an analog of the second-price rule—sale prices that render truthful bidding a dominant strategy for every bidder. The next lecture gives an affirmative answer via **Myerson's lemma**, a powerful tool in mechanism design.